

Tutorial from Quantum Optics I

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Part I

Classical theory of electromagnetic field

1 Electromagnetic field modes, Planck's law

Problem 1:

Derive from Maxwell equations in vacuum wave equation for the electric field. Show, under which conditions function $f(\vec{k} \cdot \vec{r} - \omega t)$ solves this equation.

Problem 2:

Using Fourier decomposition, show that the periodic boundary condition for the electric field ($E(\vec{r} + \vec{e}L, t) = E(\vec{r}, t)$, $\vec{e} = \vec{e}_{x,y,z}$) leads to quantization of the wave vector component values. Derive the mode density $\rho(\omega)$ at a given frequency $\omega = |\vec{k}|c$. Compare with derivation of mode density in a box with edge length L .

Problem 3:

Consider, according to Planck, the energy of an oscillator with frequency ω quantized, with values $E_n(\omega) = \hbar\omega(n + \frac{1}{2})$. Derive the average energy of one such oscillator, $g(\omega)$, assuming the Boltzmann distribution of the quantized level occupation probability, $P_n(\omega) = \frac{e^{-\frac{n\hbar\omega}{k_B T}}}{\sum_n e^{-\frac{n\hbar\omega}{k_B T}}}$. Discuss the limits $T \rightarrow 0$ (zero-point energy), $k_B T \gg \hbar\omega$ (high-temperature 'classical' limit, compare to classical harmonic oscillator), discuss the role of the temperature-independent term.

Problem 4:

Consider the total energy distribution of radiation at a given frequency and temperature $W(\omega) = \rho(\omega)g(\omega)$. Derive its (often more readily accessible) limits for high temperature ($k_B T \gg \hbar\omega$, Rayleigh) and low temperature ($k_B T \ll \hbar\omega$, Wien).

Problem 5*:

Cosmic microwave background corresponds very well to the black body radiation at $T = 2.728\text{K}$. How many photons of such radiation at the frequency of a common microwave (2.45 GHz) can be found in a volume of 1 cm^3 ? And how many at all frequencies?

(The Riemann zeta function might be useful $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty dx \frac{x^{s-1}}{e^x - 1}$, where the gamma function $\Gamma(s) = \int_0^\infty dx x^{s-1} e^{-x}$, $\Gamma(n) = (n-1)!$ and $\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s}$, so it is known that $\zeta(3) \approx 1.202$ and $\zeta(4) = \frac{\pi^4}{90}$, $\hbar = 1.05457 \cdot 10^{-34}\text{ Js}$, $k_B = 1.38065 \cdot 10^{-23}\text{ JK}^{-1}$).

2 Vector analysis

Problem 6:

Show that (for any smooth functions $\phi, \vec{A}$) holds

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

$$\nabla \times (\nabla \phi) = 0$$

Problem 7:

Verify, that for fields satisfying $\nabla \cdot \vec{A} = 0$ holds

$$\int d^3\vec{r} \quad (\nabla \times \vec{A}) \cdot (\nabla \times \vec{A}) = - \int d^3\vec{r} \quad \vec{A} \cdot \nabla^2 \vec{A}$$

Specify unmentioned requirements on analytic or asymptotic properties of $\vec{A}$.

3 Electromagnetic potentials in Coulomb gauge

Problem 8:

Let $\phi, \vec{A}$ be the scalar and vector potentials of the electromagnetic field, i.e.,

$$\begin{aligned}\vec{B} &= \nabla \times \vec{A} \\ \vec{E} &= -\frac{\partial}{\partial t} \vec{A} - \nabla \phi.\end{aligned}$$

We assume validity of Maxwell equations. In Coulomb gauge $\nabla \cdot \vec{A} = 0$

- derive source equation for scalar potential ϕ
- derive source equation for vector potential $\vec{A}$.

Problem 9:

According to Helmholtz theorem, any field can be decomposed into transverse ($\nabla \cdot \vec{E}_\perp = 0$) and longitudinal ($\nabla \times \vec{E}_\parallel = 0$), $\vec{E} = \vec{E}_\parallel + \vec{E}_\perp$.

a) Show, that the Hamiltonian

$$H = \int d\vec{r} \left(\frac{\epsilon_0}{2} \vec{E} \cdot \vec{E} + \frac{1}{2\mu_0} \vec{B} \cdot \vec{B} \right)$$

does not mix the transverse and longitudinal components of the field, and one can thus define independently $H_\perp$ and $H_\parallel$.

b) Express $H_\perp$ and $H_\parallel$ by the potentials $\phi, \vec{A}$ and discuss their meaning. For $H_\parallel$ the solution of source equation for the potential expressed by ρ will be useful, for $H_\perp$ the equality from problem 7.

Problem 10:

Show, that from the condition $\vec{q} \cdot \vec{e}_{\lambda, \vec{q}} = 0$ the polarization vectors satisfy

$$\sum_{\lambda=1,2} \mathbf{e}_{\lambda, \vec{q}}^\mu \mathbf{e}_{\lambda, \vec{q}}^{\nu*} = \delta^{\mu\nu} - \frac{\mathbf{q}^\mu \mathbf{q}^\nu}{\mathbf{q}^2}$$

instead the standard completeness relation.

4 *Variational principle, Hamilton and Lagrange formalism for electromagnetic field - optional

Problem 11:

Reminder. This problem expresses the least-action principle using Euler-Lagrange equations. Let us have a functional S in integral form

$$S = \int_{t_1}^{t_2} dt L(q_i, \dot{q}_i, t)$$

where q_i is the canonical variable and $\dot{q}_i = \frac{dq_i}{dt}$ is its time derivative and $L(q_i, \dot{q}_i, t)$ is the Lagrangian. Minimize this functional using variational calculus with fixed ends, i.e., calculate $\delta S = 0$ with zero variation at the edges. Show, that minimalization of this functional leads to Euler-Lagrange equations in the form

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0.$$

Convince yourself, that adding a functional in form of total time derivative,

$$\tilde{L} = L + \frac{d}{dt} \Psi,$$

the variational problem does not change

$$\delta \tilde{S} = \delta S,$$

and so the Euler-Lagrange equations and thus the dynamics stays the same.

Problem 12**:

Show that identifying $p = \frac{\partial \mathcal{L}}{\partial \dot{q}}$ together with the requirement that the Hamiltonian is a function of $\mathcal{H}(q, p)$ leads to the Legendre transform

$$\mathcal{H}(q, p) = p\dot{q}(p) - \mathcal{L}(q, \dot{q}(p)).$$

Start from the variation of the Lagrangian $\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \delta \dot{q}$. (Hint: $\delta(p\dot{q}) = \dot{q}\delta p + p\delta\dot{q}$).

Problem 13*:

From classical mechanics. For mass m in potential $V(x_i) = V(x, y, z)$ the second Newton's law (NL) holds

$$m\ddot{x}_i = F_i = -\frac{\partial}{\partial x_i} V.$$

i) Derive the Lagrangian $L(x_i, \dot{x}_i)$

a) by comparison of the NL with Euler-Lagrange equations

b) by direct integration of NL in the form

$$\int_{t_1}^{t_2} dt (m\ddot{x}_i - F_i) \delta x_i$$

and arranging the result into the form

$$\delta S = \int_{t_1}^{t_2} dt \delta L.$$

ii) Determine the momentum $p_i(\dot{x}_i) = \frac{\partial \mathcal{L}}{\partial \dot{x}_i}$. Inverting this relation and using the Legendre transform, derive the Hamiltonian $H(x_i, p_i)$.

iii) Formulate Hamilton's equations

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial x_i} &= -\dot{p}_i \\ \frac{\partial \mathcal{H}}{\partial p_i} &= \dot{x}_i \end{aligned}$$

Problem 14:

This problem illustrates how Lagrange and Hamilton formalisms consistently work on the example of the electromagnetic field. Start with a wave equation for the vector potential

$$-\frac{1}{\mu_0} \sum_{j=1}^3 \frac{\partial^2}{\partial x_j^2} A_i + \frac{1}{c^2 \mu_0} \frac{\partial^2 A_i}{\partial t^2} = j_i.$$

a) Comparing with Euler-Lagrange equations (see Problem 11) with canonical coordinate A_i

$$\frac{\partial \mathcal{L}}{\partial A_i} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{A}_i} - \sum_{j=1}^3 \frac{d}{dx_j} \frac{\partial \mathcal{L}}{\partial \frac{\partial A_i}{\partial x_j}} = 0$$

derive the expected form of the Lagrangian density $\mathcal{L}(A_i, \dot{A}_i, \frac{\partial A_i}{\partial x_j})$.

b) Determine the canonical momentum according to

$$P_i = \frac{\partial \mathcal{L}}{\partial \dot{A}_i}.$$

Inverting this relation, determine the Hamiltonian density by Legendre transform

$$\mathcal{H}(A_i, P_i, \frac{\partial A_i}{\partial x_j}) = P_i \dot{A}_i(P_i) - \mathcal{L}(A_i, \dot{A}_i(P_i), \frac{\partial A_i}{\partial x_j}).$$

c) Integrating over space, bring the Hamiltonian into the form corresponding to $H_\perp$ from problem 9, i.e., the energy of the transverse field. Discuss the interaction term with current $\vec{j}$.

d) Using Hamilton equations

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial A_i} &= -\dot{P}_i \\ \frac{\partial \mathcal{H}}{\partial P_i} &= \dot{A}_i \end{aligned}$$

derive the source equation for the vector potential.

5 Lagrange formalism for free electromagnetic field dynamics

Problem 15:

Lagrange density of free electromagnetic field is l is

$$l = \frac{\epsilon}{2} \frac{\partial \vec{A}}{\partial t} \cdot \frac{\partial \vec{A}}{\partial t} + \frac{1}{2\mu} \vec{A} \cdot \nabla^2 \vec{A}$$

verify, that extremalization of the action (using Euler-Lagrange equations) leads to the correct source equations for the vector potential $\vec{A}$. Consider two ways to understand the variational problem.

- a) When extremalizing the action $S = \int dr \int dt l(\vec{r}, t, \vec{A}, \frac{\partial \vec{A}}{\partial t}, \frac{\partial \vec{A}}{\partial \vec{r}})$ for a single function $\vec{A}(\vec{r}, t)$ of four variables $\vec{r}, t$, Euler-Lagrange equations are

$$-\frac{\partial l}{\partial A^i} + \frac{\partial}{\partial t} \frac{\partial l}{\partial \left(\frac{\partial A^i}{\partial t}\right)} + \sum_j \frac{\partial}{\partial r^j} \frac{\partial l}{\partial \left(\frac{\partial A^i}{\partial r^j}\right)} = 0$$

- b)*We discretize the Lagrangian $\mathcal{L}(t, \vec{A}_{\vec{r}}, \frac{\partial \vec{A}_{\vec{r}}}{\partial t}) = \sum_{\vec{r}} l(\vec{r}, t, \vec{A}, \frac{\partial \vec{A}}{\partial t}, \frac{\partial \vec{A}}{\partial \vec{r}})|_{\vec{r}}$, i.e., we seek an extremum of the action $S = \int \mathcal{L}(t, \vec{A}_{\vec{r}}, \frac{\partial \vec{A}_{\vec{r}}}{\partial t}) dt$ for a larger number of functions $A_{\vec{r}}(t)$ of a single variable t , estimating

$$\begin{aligned} \nabla^2 \vec{A} = \frac{1}{\Delta^2} [& \vec{A}_{(x+\Delta, y, z)} - 2\vec{A}_{(x, y, z)} + \vec{A}_{(x-\Delta, y, z)} \\ & + \vec{A}_{(x, y+\Delta, z)} - 2\vec{A}_{(x, y, z)} + \vec{A}_{(x, y-\Delta, z)} \\ & + \vec{A}_{(x, y, z+\Delta)} - 2\vec{A}_{(x, y, z)} + \vec{A}_{(x, y, z-\Delta)}] \end{aligned}$$

The corresponding Euler-Lagrange equations are

$$-\frac{\partial \mathcal{L}}{\partial A_{\vec{r}}^i} + \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \left(\frac{\partial A_{\vec{r}}^i}{\partial t}\right)} = 0$$

Further verify, that Coulomb gauge is only a boundary condition of the Euler-Lagrange equations (i.e., selects a solution of the Euler-Lagrange equations), and does not represent an additional constraint of the variation problem.

Problem 16:

Start with the transverse field Hamiltonian in the form (result of problem 9)

$$H_{\perp} = \int d\vec{r} \left(\frac{\epsilon_0}{2} \frac{\partial A(r, t)}{\partial t} \frac{\partial A(r, t)}{\partial t} - \frac{1}{2\mu_0} A(r, t) \nabla^2 A(r, t) \right).$$

Derive the Hamiltonian in Fourier space, that is, for

$$A(\vec{r}, t) = \sum_{\lambda, \vec{q}} f_{\lambda \vec{q}}(\vec{r}) A_{\lambda \vec{q}}(t),$$

where

$$f_{\lambda \vec{q}}(\vec{r}) = \frac{e_{\lambda \vec{q}}}{\sqrt{V}} e^{i\vec{q} \cdot \vec{r}} \text{ a } A_{\lambda \vec{q}}(t) = \frac{1}{\sqrt{V}} A_{\lambda}(\vec{q}).$$

It will be useful, that

$$\int d\vec{r} f_{\lambda \vec{q}}(\vec{r}) f_{\mu \vec{k}}(\vec{r}) = \delta_{\lambda \mu} \delta(\vec{q} + \vec{k}).$$

Problem 17:

Using Hamiltonian of the free electromagnetic field

$$H = \frac{1}{2\epsilon} \sum_{\lambda, \vec{k}} \left(P_{\lambda, -\vec{k}} P_{\lambda, \vec{k}} + \frac{k^2 \epsilon}{\mu} A_{\lambda, -\vec{k}} A_{\lambda, \vec{k}} \right) \quad (1)$$

calculate the classical Poisson brackets $\{H, P_{\lambda, \vec{k}}\}_P, \{H, A_{\lambda, \vec{k}}\}_P$. Poisson brackets are

$$\{B, C\}_P \equiv \sum_{\lambda, \vec{k}} \frac{\partial B}{\partial A_{\lambda, \vec{k}}} \frac{\partial C}{\partial P_{\lambda, \vec{k}}} - \frac{\partial C}{\partial A_{\lambda, \vec{k}}} \frac{\partial B}{\partial P_{\lambda, \vec{k}}}$$

Part II**Canonical quantization of the electromagnetic field****6 Energy and momentum seen by the photons****Problem 1:**

Which quantum-mechanical commutators correspond by the correspondence principle to the Poisson brackets in problem 1.17?

Problem 2:

Show, that the Hamiltonian (1) from problem 1.16 after second quantization

$$A_{\lambda, \vec{k}} = \sqrt{\frac{\hbar}{2\epsilon\omega_{\vec{k}}}} (a_{\lambda, -\vec{k}}^{\dagger} + a_{\lambda, \vec{k}})$$

$$P_{\lambda, \vec{k}} = i\epsilon\omega_{\vec{k}} \sqrt{\frac{\hbar}{2\epsilon\omega_{\vec{k}}}} (a_{\lambda, \vec{k}}^{\dagger} - a_{\lambda, -\vec{k}})$$

obtains the form

$$H = \sum_{\lambda, \vec{k}} \hbar\omega_{\vec{k}} \left(a_{\lambda, \vec{k}}^{\dagger} a_{\lambda, \vec{k}} + \frac{1}{2} \right)$$

Problem 3:

Discuss Heisenberg and Schrödinger picture of the time evolution of the free electromagnetic field.

Problem 4:

Show, that the momentum (Poynting vector) of the electromagnetic field

$$\frac{\vec{S}}{c^2} = \epsilon \int d^3\vec{r} \quad \vec{E}(\vec{r}) \times \vec{B}(\vec{r})$$

can be interpreted as the momentum of individual photons.

Problem 5:

Calculating the following commutators (time evolution in Heisenberg picture)

$$[E^\mu(r, t), B^\nu(r', t')]$$

decide, which observations of the electric and magnetic components of the light wave are incompatible. (Hint: bear Problem 1.10 in mind).

7 Creation operator algebra**Problem 7:**

Evaluate the following commutators

- $[a, a^\dagger]$
- $[a, a^{\dagger 2}]$

Problem 7:

Prove, that, $aa^{\dagger N} = a^{\dagger N}a + Na^{\dagger(N-1)}$.

Problem 8:

Using mathematical induction or by a direct calculation, show that

$$[(a \pm a^\dagger)^n, a] = \mp n(a \pm a^\dagger)^{n-1}$$

and use the result for evaluation of the commutator

$$[e^{\xi(a^\dagger - a)}, a]$$

Problem 9:

Using superoperator notation $L_\zeta \hat{X} \equiv \frac{1}{2}[\hat{X}, \bar{\zeta}a^2 - \zeta a^{\dagger 2}]$, convince yourself that

$$L_\zeta a = -\zeta a^\dagger$$

$$L_\zeta a^\dagger = -\bar{\zeta} a$$

and prove the relation

$$e^{L_\zeta} a = a \cosh |\zeta| - a^\dagger \zeta \sinh |\zeta| / |\zeta|$$

Příklad 10:

Interpreting super-operator exponential of the commutator $L_A \hat{X} \equiv [A, \hat{X}]$ in Hilbert space, derive the Hadamard lemma

$$e^A B e^{-A} = B + \frac{1}{1!}[A, B] + \frac{1}{2!}[A, [A, B]] + \frac{1}{3!}[A, [A, [A, B]]] + \dots$$

In its light discuss Heisenberg picture of creation and annihilation operator of the electromagnetic field.

Problem 11*:

Using Hadamard lemma, derive the Baker-Campbell-Hausdorff formula in the form

$$e^A e^B = e^B e^A e^{[A,B]}, \text{ pokud } [A, [A, B]] = [B, [A, B]] = 0.$$

Discuss the implications for creation operator algebra, for example normalize the wave function $|\psi\rangle = e^{\alpha a^\dagger} |\phi\rangle$. Using the Trotter formula

$$e^{A+B} = \lim_{n \rightarrow \infty} \left(e^{A/n} e^{B/n} \right)^n$$

further verify another form of the Baker-Campbell-Hausdorff formula

$$e^{A+B} = e^B e^A e^{[A,B]/2}.$$

Part III**Pure photon states****8 (Fock) photon number states****Problem 1:**

Normalize the photon state with number N of photons in mode $\lambda, \vec{k}$. I.e., find $\mathcal{N}$ in

$$|N_{\lambda, \vec{k}}\rangle = \mathcal{N} \left(a_{\lambda, \vec{k}}^\dagger \right)^{N_{\lambda, \vec{k}}} |\phi\rangle$$

such that $\langle N_{\lambda, \vec{k}} | N_{\lambda, \vec{k}} \rangle = 1$.

Problem 2:

Evaluate the expectation value of the electric field

$$\vec{E}_T(\vec{r}) = -i \sum_{\lambda, \vec{k}} \sqrt{\frac{\hbar \omega_{\vec{k}}}{2\epsilon}} \left(\mathbf{e}_{\lambda, -\vec{k}} e^{-i\vec{k} \cdot \vec{r}} a_{\lambda, \vec{k}}^\dagger - \mathbf{e}_{\lambda, \vec{k}} e^{i\vec{k} \cdot \vec{r}} a_{\lambda, \vec{k}} \right)$$

in the state

$$|\psi\rangle = c_n |N_{\lambda, \vec{k}}\rangle + c_{n-1} |(N-1)_{\lambda, \vec{k}}\rangle$$

Discuss its time evolution as well.

Problem 3*:

Discuss the distribution of electric field values in electromagnetic vacuum states considering all modes of the electric field. Take into account the role of the finite spatial resolution of measurement (e.g., by convolution of the E field with a Gaussian measurement function).

9 Coherent states**Problem 4:**

In mode $\lambda \vec{k}$ the coherent states are defined as

$$|\alpha_{\lambda \vec{k}}\rangle \equiv e^{-\frac{1}{2}|\alpha_{\lambda \vec{k}}|^2 + \alpha_{\lambda \vec{k}} a_{\lambda \vec{k}}^\dagger} |\phi\rangle,$$

where $\alpha = |\alpha| e^{i\theta}$ is a complex parameter. evaluate the scalar product $\langle \beta_{\lambda \vec{k}} | \alpha_{\lambda \vec{k}} \rangle$.

Problem 5:

for coherent states holds $|\alpha\rangle = D_\alpha |\phi\rangle$. Show, that for the shift operator $D_\alpha = e^{\alpha a^\dagger - \alpha^* a}$ holds

$$D_\alpha^\dagger a D_\alpha = a + \alpha.$$

Problem 6:

Show, that the coherent state definition in problem 4 is equivalent with the shift in problem 5, and that it is equivalent with the definition as an eigenstate of the annihilation operator:

$$a|\alpha\rangle = \alpha|\alpha\rangle.$$

Does an eigenstate of the creation operator exist? Prove.

Problem 7*:

Calculate the energy and fluctuation of the electromagnetic field in the coherent state.

10 Squeezed states***Problem 8:**

Light is called quadrature-squeezed if the electric field fluctuations are smaller than those of the vacuum field. The squeezing operator is in this case

$$S_\zeta = e^{\frac{1}{2}\zeta^* a^2 - \frac{1}{2}\zeta a^{\dagger 2}},$$

where $\zeta = se^{i\theta}$ is the squeeze parameter.

Show, that

$$S_\zeta^\dagger a S_\zeta = a \cosh(s) - a^\dagger e^{i\theta} \sinh(s).$$

Problem 9:

Consider squeezed vacuum $S_\zeta|\emptyset\rangle$. Calculate a) mean number of photons, b)* photon number fluctuation, c) mean value of the electric field and d)* electric field fluctuations.

Problem 10*:

Discuss the dependence of the fluctuations (i.e., the variance) of the electric or magnetic field in a squeezed state on the phase of the field $\varphi \equiv \mathbf{q} \cdot \mathbf{r}$.

Příklad 11:**

Show, that the squeezed vacuum $S_\zeta|\emptyset\rangle$, is a superposition of states with even number of photons:

$$|\zeta\rangle = S_\zeta|\emptyset\rangle = \frac{1}{\sqrt{\text{sech}s}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{n!} \left[-\frac{1}{2}e^{i\theta} \tanh s \right]^n |2n\rangle.$$

Problem 12: Squeezed state generation by parametric down-conversion

Squeezed states can be generated by the process of degenerate down-conversion, i.e., a nonlinear process in which one photon of frequency $2\omega_0$ is converted into two photons of frequency ω_0 . Such experiment was realized by Wu et al. (PRL 57, 2520 (1986)). The $\text{MgO}:\text{LiNbO}_3$ nonlinear crystal, where the process takes place, was placed in an optical parametric oscillator. The generating field of the frequency-doubled YAG laser is then strong enough to be described semi-classically, using a complex amplitude ε . The mode, in which the photon pairs are generated, can then be described by the interaction Hamiltonian

$$H = \hbar\omega_0 a^\dagger a - i\hbar g (a^{\dagger 2} \varepsilon e^{-i2\omega_0 t} - a^2 \varepsilon^* e^{i2\omega_0 t}), \quad (2)$$

where g is a constant scaling the interaction (depends on the crystal nonlinearity $\chi^{(2)}$). The terms with $a^{\dagger 2}$ reflect the generation of photon pairs, the generation is proportional to the pump field amplitude ε .

determine the time evolution of the operator $a(t)$ and $a^\dagger(t)$ in Heisenberg picture, i.e., solve

$$\dot{a}(t) = \frac{i}{\hbar} [H, a(t)] \quad (3)$$

with the Hamiltonian above. Hint: $a(t) = e^{-i\omega_0 t} a^I(t)$ and solve as two coupled equations for $a(t)$ and $a^\dagger(t)$. Show, that the solution is:

$$a(t) = a(0) \cosh(2g|\varepsilon|t) e^{-i\omega_0 t} - \frac{\varepsilon}{|\varepsilon|} a^\dagger(0) \sinh(2g|\varepsilon|t) e^{-i\omega_0 t}. \quad (4)$$

Assuming the generated mode in time 0 in vacuum state, $\langle \emptyset | a^\dagger(0) = 0$, $a(0)|\emptyset\rangle = 0$, derive the expectation value $\langle \emptyset | E(t) | \emptyset \rangle$ and variance $\langle \emptyset | \Delta E(t)^2 | \emptyset \rangle$, again in Heisenberg picture. Compare with the result of problem 10 and by comparison of the results identify the squeezing parameter. Is the light amplitude- or phase-squeezed? Consider interference with a reference wave, so-called local oscillator (LO), Sketch the dependence of the detected intensity of the interfering fields on the phase of the LO.

Part IV

Electromagnetic field in matter

11 Mixed states*

Problem 1**:

Verify, that for operators $\hat{A}$, $\hat{B}$, $\hat{C}$ on finite-dimensional Hilbert space holds the cyclic invariance of the trace

$$\text{Tr}(\hat{A}\hat{B}\hat{C}) = \text{Tr}(\hat{C}\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{C}\hat{A})$$

Discuss possible extension to infinite-dimension spaces.

Problem 2*:

Prove: For pure state $\text{Tr}(\hat{\rho}^2) = 1$; otherwise $\text{Tr}(\hat{\rho}^2) < 1$.

Problem 3:

Take density matrix

$$\hat{\rho} = \frac{1}{2} \begin{pmatrix} 1 & ge^{-i\varphi} \\ ge^{i\varphi} & 1 \end{pmatrix}$$

where $g, \varphi \in R^+$. Verify, that this is a density matrix of a pure state for $g = 1$ and of a mixed state for $g < 1$.

Problem 4:

Verify, that in the eigenstate basis of the Hamiltonian the density matrix is time-independent.

12 Thermal blackbody radiation

Problem 5:

Derive Bose-Einstein distribution of the occupation numbers

$$n_{\lambda\mathbf{q}} \equiv \text{Tr} \left\{ \hat{\rho}_{\text{eq}} a_{\lambda\mathbf{q}}^\dagger a_{\lambda\mathbf{q}} \right\} = \frac{1}{e^{\beta\hbar\omega} - 1}$$

for equilibrium radiation density matrix $\hat{\rho}_{\text{eq}} \equiv e^{-\beta \sum_{\lambda\mathbf{q}} \hbar\omega_{\mathbf{q}} a_{\lambda\mathbf{q}}^\dagger a_{\lambda\mathbf{q}}} / \text{Tr} \{ e^{-\beta \sum_{\lambda\mathbf{q}} \hbar\omega_{\mathbf{q}} a_{\lambda\mathbf{q}}^\dagger a_{\lambda\mathbf{q}}} \}$.

Problem 6:

Calculate the energy of the blackbody radiation from problem 5

$$E_T \equiv \text{Tr} \left\{ \hat{\rho}_{\text{eq}} \hat{H}_T \right\} = \sum_{\lambda\mathbf{q}} \hbar\omega_{\mathbf{q}} n_{\lambda\mathbf{q}}.$$

13 Dipole interaction of molecules with electromagnetic field

Problem 7:

Verify the validity of the following operator identity

$$\frac{1}{\hat{A} - \hat{B}} = \frac{1}{\hat{A}} + \frac{1}{\hat{A}} \hat{B} \frac{1}{\hat{A} - \hat{B}}$$

Problem 8:

Show, that in Coulomb gauge holds

$$\vec{p} \cdot \vec{A} = \vec{A} \cdot \vec{p}$$

Problem 9:

The rate of spontaneous transitions in the hydrogen atom can be calculated by Fermi golden rule

$$\mathcal{P}_{a \rightarrow} = \sum_b \frac{2\pi}{\hbar} \delta(E_a - E_b) |\langle a | H_{\text{int}} | b \rangle|^2$$

Define the multipole expansion of one-photon transitions to the ground state $1s$ of the hydrogen atom, classify the type of the transition for $2s \rightarrow 1s$, $2p \rightarrow 1s$, $3d \rightarrow 1s$ and roughly or exactly compare the rate of these transitions. The wave functions of the hydrogen atom eigenstates (for $m = 0$) are

$$\begin{aligned} \psi_{1s} &= \frac{1}{\sqrt{\pi R_B^3}} e^{-r/R_B} \\ \psi_{2s} &= \frac{1}{\sqrt{8\pi R_B^3}} \left(1 - \frac{r}{2R_B}\right) e^{-r/2R_B} \\ \psi_{2p} &= \frac{1}{4\sqrt{2\pi R_B^3}} (r/R_B) e^{-r/2R_B} \cos \theta. \\ \psi_{3d} &= \frac{1}{81\sqrt{6\pi R_B^3}} (r/R_B)^2 e^{-r/3R_B} (3 \cos^2 \theta - 1). \end{aligned}$$

where $R_B = 5.3 \times 10^{-11} m$ is the Bohr radius. The largest wavelength of $1s$ ionizing radiation is 364.56 nm.

Problem 10:

For interaction Hamiltonian

$$\hat{H}_I = \sum_i \frac{e_i}{m} \vec{p}_i \cdot \vec{A}(r_i, t)$$

evaluate the element

$$\langle \psi_f | \hat{H}_I | \psi_l \rangle$$

where $|\psi_{l,f}\rangle$ are the eigenstates of the molecular Hamiltonian $\hat{H}_0 = \sum_i \left(\frac{p_i^2}{2m} + \sum_{j \neq i} \frac{e_i e_j}{2\pi\epsilon |r_i - r_j|} \right)$

$$\hat{H}_0 |\psi_f\rangle = E_f |\psi_f\rangle$$

Express the result using transition dipole moment

$$\vec{d}_{fl} = \sum_i e_i \langle \psi_f | \vec{r} | \psi_l \rangle$$

Problem 11:

Consider two-level system a (eigenstates $|a : 0\rangle$, $|a : 1\rangle$) with transition of dipole character $\hat{d} = |0\rangle d_{01} \langle 1| + h.c.$, and electromagnetic field f in mode of frequency $\omega \approx \omega_{10}$. Compare the probabilities of the following transitions $|i\rangle \rightarrow |f\rangle$:

- $|i\rangle = |a : 1\rangle |f : N\rangle, |f\rangle = |a : 0\rangle |f : N \pm 1\rangle$
- $|i\rangle = |a : 0\rangle |f : N\rangle, |f\rangle = |a : 1\rangle |f : N \pm 1\rangle$
- $|i\rangle = |a : 0\rangle |f : \alpha\rangle, |f\rangle = |a : 1\rangle |f : \beta\rangle$
- $|i\rangle = |a : 1\rangle |f : \alpha\rangle, |f\rangle = |a : 0\rangle |f : \beta\rangle$

Part V

Two-level transition dynamics

14 Spontaneous emission

Problem 1:

Calculate the lifetime of the 2p state of hydrogen by evaluating

$$\tau^{-1} = \frac{(E_{2p} - E_{1s})^3 d_{1s,2p}^2}{3\pi\epsilon_0 \hbar^4 c^3}.$$

The energy of hydrogen ground state level is $E_{1s} = -13.5\text{eV} = -21.6 \times 10^{-19}\text{J}$ and holds $E_{2p} = E_{1s}/4$. Vacuum permittivity is $\epsilon_0 = 8.854 \times 10^{-12}\text{F/m}$. The wave functions are given as

$$\begin{aligned}\psi_{1s} &= \frac{1}{\sqrt{\pi R_B^3}} e^{-r/R_B} \\ \psi_{2p} &= \frac{1}{4\sqrt{2\pi R_B^3}} (r/R_B) e^{-r/2R_B} \cos \theta.\end{aligned}$$

where $R_B = 5.3 \times 10^{-11}\text{m}$ is the Bohr radius.

Problem 2:

Compare the natural linewidth (corresponding to the lifetime in Problem 1) with thermal (Doppler) broadening

$$\omega_{th} = \frac{E_n - E_m}{\hbar} \sqrt{\frac{2k_B T}{mc^2}} \text{ at room temperature } T = 300\text{K}.$$

You will need: hydrogen atom mass $m = 1.67 \times 10^{-27}\text{kg}$, Boltzmann constant $1.38 \times 10^{-23}\text{JK}^{-1}$.

Problem 3:

Verify, that for the set of (Einstein) kinetic equations

$$\begin{aligned}\frac{d\rho_{11}}{dt} &= 2\gamma_2\rho_{22} - 2\gamma_1\rho_{11} \\ \frac{d\rho_{22}}{dt} &= 2\gamma_1\rho_{11} - 2\gamma_2\rho_{22}\end{aligned}$$

describing the non-resonant dynamics of a two-level transition holds in equilibrium

$$\frac{\rho_{11}}{\rho_{22}} = \frac{\gamma_2}{\gamma_1}$$

Further express equilibrium ρ_{11}, ρ_{22} taking into account detailed balance.

Problem 4:

Using a substitution $\tilde{\psi}_2 = e^{i\omega_0 t} \psi_2$ convert the Schrödinger equation for resonantly driven two-level system

$$\frac{d}{dt} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = -\frac{i}{\hbar} \begin{pmatrix} \mathcal{E}_1 & -d_{12}E_0 e^{i\omega_0 t} \\ -d_{21}E_0 e^{-i\omega_0 t} & \mathcal{E}_2 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

to an equation with constant coefficients and find its fundamental system.

Part VI

Test example

Always take into account a single mode of light only.

Problem 1:

Similar as for coherent states, the time evolution of the set of squeezed states is a (group) permutation. Prove this and find the permutation. Does this stay valid for a Hamiltonian which is not normally ordered, i.e., $H = \frac{\hbar\omega}{2}(aa^\dagger + a^\dagger a)$?

Problem 2:

Evaluate

$$\langle \alpha | a a a^\dagger a^\dagger a^\dagger a a | \alpha \rangle.$$

for coherent state $|\alpha\rangle$.

Problem 3:

Can we prescribe a wave function Ψ also just by its function

$$f_\Psi(\alpha) \equiv \langle \alpha | \Psi \rangle$$

on the set (or part of the set) of coherent states $|\alpha\rangle$?

Solution:

1. The time evolution of the squeezed state is

$$\begin{aligned} |\zeta(t)\rangle &\equiv e^{\frac{-i}{\hbar}\hat{H}t} e^{\frac{1}{2}\bar{\zeta}a^2 - \frac{1}{2}\zeta a^{\dagger 2}} |\phi\rangle = e^{\frac{-i}{\hbar}\hat{H}t} e^{\frac{1}{2}\bar{\zeta}a^2 - \frac{1}{2}\zeta a^{\dagger 2}} e^{\frac{i}{\hbar}\hat{H}t} |\phi\rangle = \\ &= e^{\frac{1}{2}\bar{\zeta} e^{\frac{-i}{\hbar}\hat{H}t} a^2 e^{\frac{i}{\hbar}\hat{H}t} - \frac{1}{2}\zeta e^{\frac{-i}{\hbar}\hat{H}t} a^{\dagger 2} e^{\frac{i}{\hbar}\hat{H}t}} |\phi\rangle = e^{\frac{1}{2}e^{2i\omega t}\bar{\zeta}a^2 - \frac{1}{2}e^{-2i\omega t}\zeta a^{\dagger 2}} |\phi\rangle = |\zeta e^{-2i\omega t}\rangle \end{aligned}$$

i.e., $\zeta(t) = \zeta e^{-2i\omega t}$; q.e.d. The shift of the ground state of the mode oscillator (in the non-normal ordering) leads to only physically unimportant phase factor $e^{-i\omega t/2}$ (state is determined by the ray in Hilbert state).

2. Directly evaluating

$$\begin{aligned} \langle \alpha | a^2 a^{\dagger 3} a^2 | \alpha \rangle &= \alpha^2 \langle \alpha | a^2 a^{\dagger 3} | \alpha \rangle = \alpha^2 \langle \phi | (a + \alpha)^2 (a^\dagger + \bar{\alpha})^3 | \phi \rangle = \\ &= \alpha^2 (3\bar{\alpha} \langle \phi | a^2 a^{\dagger 2} | \phi \rangle + 2\alpha 3\bar{\alpha}^2 \langle \phi | a a^\dagger | \phi \rangle + \alpha^2 \bar{\alpha}^3 \langle \phi | \phi \rangle) \\ &= \alpha^2 \bar{\alpha} (6 + 6|\alpha|^2 + |\alpha|^4) \end{aligned}$$

3. Ψ can be expanded in the Fock photon number state basis

$$\Psi = \sum_{N=0}^{\infty} c_N |N\rangle$$

if we find from $f_\Psi(\alpha)$ coefficients c_N , the problem is solved. This is not difficult, see

$$f_\Psi(\alpha) \equiv \langle \alpha | \Psi \rangle = \sum_{N=0}^{\infty} \sum_{N'=0}^{\infty} \langle N' | \frac{e^{-\frac{1}{2}|\alpha|^2} \bar{\alpha}^{N'}}{\sqrt{N'!}} c_N | N \rangle = \sum_{N=0}^{\infty} \frac{e^{-\frac{1}{2}|\alpha|^2} \bar{\alpha}^N}{\sqrt{N!}} c_N$$

It is thus sufficient to restrict ourselves to a small interval around $\alpha = 0$ on the real axis and find

$$c_N = \frac{1}{\sqrt{N!}} \frac{d^N}{d\alpha^N} e^{\frac{1}{2}\alpha^2} f_\Psi(\alpha) \Big|_{\alpha=0}$$